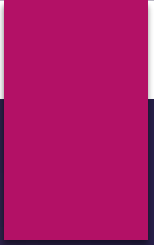


Terraform Cloud & Spacelift

ENTERPRISE TERRAFORM AUTOMATION, GOVERNANCE & GITOPS WORKFLOWS

Learning Objectives

1. Understand Terraform Cloud and Spacelift
2. Compare local Terraform vs managed platforms
3. Explain remote state and remote execution
4. Configure GitOps-based Terraform workflows
5. Understand Policy as Code concepts
6. Differentiate Terraform Cloud and Spacelift
7. Build secure enterprise IaC pipelines



INTRODUCTION TO TERRAFORM AUTOMATION PLATFORMS

ENTERPRISE INFRASTRUCTURE AUTOMATION PLATFORMS

Managing Terraform at Scale

Enterprise Requirements

- ▶ Secure state management
- ▶ Team collaboration
- ▶ Policy enforcement
- ▶ CI/CD integration
- ▶ Drift detection
- ▶ Audit logging

Managing Terraform at Scale

Problems with Local Terraform Workflows

- ▶ Local state management risks
- ▶ No centralized governance
- ▶ Inconsistent execution environments
- ▶ Difficult collaboration
- ▶ Limited auditability
- ▶ Credential management challenges
- ▶ Manual deployment workflows

What is Terraform Cloud?

- ▶ HashiCorp **Terraform Cloud** is a managed platform by HashiCorp that helps teams automate, collaborate, secure, and manage Infrastructure as Code (IaC) workflows using Terraform.
- ▶ Terraform Cloud was rebranded to **HCP Terraform** as part of HashiCorp's transition into the HashiCorp Cloud Platform (HCP)

It provides a centralized environment for:

- ▶ Running Terraform remotely
 - ▶ Managing Terraform state securely
 - ▶ Collaborating across teams
 - ▶ Enforcing policies and governance
 - ▶ Integrating CI/CD workflows
 - ▶ Managing variables and secrets
 - ▶ Using private module registries
- ▶ Instead of running Terraform only from a local machine, Terraform Cloud enables teams to manage infrastructure in a controlled and scalable way.

Key Features of Terraform Cloud

- ▶ **Remote State Management:** Terraform state is stored securely in Terraform Cloud instead of locally.
- ▶ **Remote Terraform Runs:** Terraform commands run inside Terraform Cloud workers.
- ▶ **VCS Integration:** Integrates with GitHub, GitLab, Bitbucket, Azure DevOps
- ▶ **Workspace Management:** A workspace is an isolated Terraform working environment.
- ▶ **Variables & Secrets Management:** securely stores , Environment variables, Terraform variables, API tokens, Cloud credentials.
- ▶ **Policy as Code:** Terraform Cloud supports governance using, Sentinel policies, OPA integrations.
- ▶ **Private Module Registry:** Organizations can publish reusable Terraform modules privately

Terraform Cloud Pricing Tiers

Tier	Features
Free	Small teams, remote state, VCS integration
Team & Governance	RBAC, policy enforcement
Business	Advanced governance & SSO

Terraform Cloud Workflow

- ▶ Create organization
- ▶ Create workspace
- ▶ Connect GitHub repository
- ▶ Configure variables
- ▶ Trigger remote plan/apply

When to Use Terraform Cloud

- ▶ Terraform-focused teams
- ▶ Smaller organizations
- ▶ Simple GitOps workflows
- ▶ Native Terraform ecosystem users

What is Spacelift?

- ▶ Spacelift is an Infrastructure-as-Code (IaC) orchestration platform designed to automate, secure, and govern infrastructure deployments at scale. It works with tools like Terraform, OpenTofu, Pulumi, Kubernetes, CloudFormation, and Ansible.

Think of Spacelift as:

- ▶ A CI/CD platform specialized for infrastructure
- ▶ A governance and policy engine for IaC
- ▶ A collaboration platform for infrastructure teams
- ▶ A centralized control plane for Terraform and cloud deployments

Key Features of Spacelift

- ▶ **Terraform & Multi-IaC Support:** Spacelift supports Terraform, OpenTofu, Pulumi, Kubernetes etc.
- ▶ **GitOps Workflows:** Infrastructure changes are triggered from Git repositories.
- ▶ **Policy as Code (OPA):** Spacelift integrates with Open Policy Agent (OPA).
- ▶ **Drift Detection:** Spacelift continuously monitors infrastructure for drift.
- ▶ **Dynamic Cloud Credentials:** Instead of storing static secrets, Spacelift supports short-lived credentials using OIDC integrations.
- ▶ **Stack Management:** A stack is the main Spacelift deployment unit.

SPACELIFT Pricing Tiers

Plan	Starting Price	Best For	Key Features
Free	\$0	Individuals & small teams	2 users, 1 API key, public workers, Git integrations
Starter	From \$399/month	Growing teams	10 users, Policy as Code, private module registry
Starter+	Custom quote	Teams needing private workers	Unlimited users, drift detection, 1 private worker
Business	Custom quote	Enterprise automation	Advanced scheduling, private provider registry
Enterprise	Custom quote	Large enterprises	Audit trails, SAML SSO, self-hosted option

TERRAFORM CLOUD VS SPACELIFT

Feature	Terraform Cloud	Spacelift
Built By	HashiCorp	Spacelift
Primary Focus	Terraform	Multi-IaC
Policy Engine	Sentinel, OPA	OPA
Drift Detection	Limited	Advanced
Multi-IaC Support	Limited	Extensive
Customization	Moderate	High

Traditional CI/CD Vrs Terraform Cloud Vrs Spacelift

Feature / Capability	Traditional CI/CD (GitHub Actions, Jenkins, GitLab CI, etc.)	Terraform Cloud	Spacelift
Primary Purpose	General software CI/CD automation	Terraform automation platform	Enterprise IaC orchestration platform
Built For	Applications + Infrastructure	Terraform workflows	Multi-IaC workflows
IaC Awareness	Limited (manual scripting required)	Native Terraform-aware	Deep IaC-aware orchestration
Supported IaC Tools	Any (manual integration)	Mainly Terraform	Terraform, OpenTofu, Pulumi, Kubernetes, Ansible, CloudFormation
Terraform State Management	Manual setup required	Built-in remote state management	Built-in state management integration
State Locking	Requires backend configuration	Built-in	Built-in

Traditional CI/CD Vrs Terraform Cloud Vrs Spacelift

Feature / Capability	Traditional CI/CD (GitHub Actions, Jenkins, GitLab CI, etc.)	Terraform Cloud	Spacelift
Remote Terraform Execution	Self-managed runners	Managed remote runs	Managed workers/runners
VCS Integration	Yes	Native integration	Native integration
Pull Request Workflows	Requires manual setup	Built-in PR workflows	Built-in GitOps workflows
GitOps Support	Manual implementation	Supported	Strong GitOps-first design
Drift Detection	Manual scripting required	Limited/basic	Advanced built-in drift detection
Policy as Code	Requires external tools	Sentinel	OPA (Open Policy Agent)
Approval Workflows	Manual/custom setup	Built-in	Built-in advanced workflows
Ideal Organization Type	Any DevOps team	Terraform-centric organizations	Enterprise platform engineering teams

Quick Summary

Platform	Best Use Case
Traditional CI/CD	General application + infrastructure automation
Terraform Cloud	Teams focused primarily on Terraform workflows
Spacelift	Enterprises managing large-scale multi-IaC environments with strong governance needs